Unified Camera + LiDAR + Radar | Illustrated Research Synthesis

GEMSTAR Task 23 system-level synthesis: data provenance, coordinate/timing conventions, interface integration, evidence readiness and research gaps.

Evidence source: uploaded Task_23_Final_Research.zip. Figures and tables come from archived assets. This document explains established findings separately from assumptions and future validation.

Executive decision

Task 23 demonstrates sensor-specific data availability, static simulator calibration, reusable preprocessing and synthetic three-input software compatibility. It does not establish a recorded-OSI same-run end-to-end three-sensor pipeline, independent acquisition synchronization, measured Camera reprojection, ground-truth tracking or energy efficiency. This report integrates the validated evidence without conflating those milestones.

Full pipeline: purpose, data interfaces, and boundaries

Input	Validated preprocessing	Output / next stage
Camera 640x480 RGB	ROI validation, grayscale, Canny, luminance, contrast, Laplacian, saturation	SensorMeasurement: Camera-local features; no camera object IDs
LiDAR reflection arrays	Positive-return availability, validity/status, local-to-vehicle coordinate adapter	SensorMeasurement: LiDAR quality and optional 3D points pending end-to-end wiring
Radar reflection TOF/strength	Positive finite validation, range c*TOF/2, separate zero-rotation geometry	SensorMeasurement: Radar range/strength, geometry pending end-to-end wiring
SensorView timestamp	Integer nanosecond validation; missing/stale/future detection	Preliminary bundle policy, not independent synchronization

System subtask synthesis, in historical order

23.2-3 - OSI discovery and Camera payload interpretation

Initial schema/Camera decoding checks established how to interpret 3-channel images and reject competing color-layout hypotheses; MATLAB and Python outputs support reuse of archived selected frames.

23.4-5 - LiDAR and Radar reconstruction plus exploratory spatial analysis

Source-driven scan geometry, zero-rotation coordinate conversion, Radar TOF/range reconstruction, MATLAB cross-checks, and tentative radar-lidar overlap/association diagnostics were produced. These are not object GT accuracy metrics.

23.6-10 - Controlled geometric calibration

Camera, LiDAR and Radar used controlled cube displacement and static map/configuration workflows. Several setups use different mounting parents and separate runs, so measured values cannot automatically be fused across tasks.

23.11 - LiDAR weather and availability

40,802 frames across Clear/Fog100/Fog20 quantified positive return availability. Results were static, with Fog100 the same availability as Clear and Fog20 zero positive returns in that scene; Python scoring was profiled on the development PC.

23.12 - Camera controlled weather

Separate Clear/Fog100/Fog20 Camera recordings and near-time static images, five quality features and MATLAB central-ROI agreement; image quality is not detection correctness.

23.13 - Radar static OSI range serialization

Four central rays verified simulated geometric displacement versus OSI TOF-decoded displacement within the predefined diagnostic tolerance; physical Radar sensing and dynamic objects untested.

23.14 - Coordinate and temporal readiness audit

Nominal Camera intrinsics agree with two cube-width samples; configured 3D sensor translations calculated; 15/15 sampled Radar-LiDAR records share SensorView timestamp equal to GT and 9/9 sampled intervals increase. Independent sensor acquisition synchronization not established.

23.15 - Unified software interface and synthetic integration

Six new modules implement a common SensorMeasurement container, Camera/LiDAR/Radar adapters, zero-rotation coordinate adapter, timestamp bundle checks. Synthetic tests: interface 14/14, Camera 7/7, LiDAR 9/9, Radar 13/13, coordinates 7/7, timestamps 12/12, combined 12/12.

23.16 - Pending final archive acceptance

Research explanation and figure-to-source register can now be included, but scientific gates should be explicitly classified as complete/limited/deferred before sign-off.

Geometry: correctly interpreting mount differences

Sensor	Original named mount	Configured location (m)
Camera	SensorPlatform_1	(0.70, 0, 1.60)
LiDAR	SensorPlatform_1	(2.60, 0, 0.70)
Radar	SensorPlatform_1	(2.00, 0, 0.80)

Original zero-rotation relative distances: Camera-LiDAR 2.102380 m, Camera-Radar 1.526434 m, Radar-LiDAR 0.608276 m. In Task 23.12 the LiDAR configuration instead had no named mount and was placed at (19.40,-26.50,2.50) m while Camera/Radar were vehicle-mounted. Do not treat those recordings as a common physical reference frame.

Time: what 15 equal timestamps actually mean

In 15 audited Radar-LiDAR records the SensorView timestamp matched GroundTruth and Radar/LiDAR belonged to a shared SensorView container. No independent individual acquisition timestamp was available. Frame steps in the sampled Radar-LiDAR records span 27.894-38.361 ms, mean 32.803 ms. Camera intervals from separate runs are Clear 36.865, Fog100 39.425 and Fog20 33.509 ms. None of these numbers measures the inter-sensor acquisition

offset.

Software validation acceptance vs research validation

Area	Status	What is still missing
Sensor parsing / synthetic adapters	Supported in tested cases	Combined recorded-OSI smoke test and broader robustness
Configured extrinsic vectors	Mathematically consistent	Effective 6-DOF runtime mounts and optical axis
Shared container time	Sampled Radar/LiDAR PASS	Independent Camera/LiDAR/Radar capture-time evidence
Camera reprojection	Not established	Matched 3D/2D target correspondences and pixel-error summary
Object association/EKF	Not established	GT tracking and dynamic scenes in Task 24
Energy efficiency	Not established	Matched CPU/FPGA power-energy and latency measurements later

System-level visual evidence from the archive

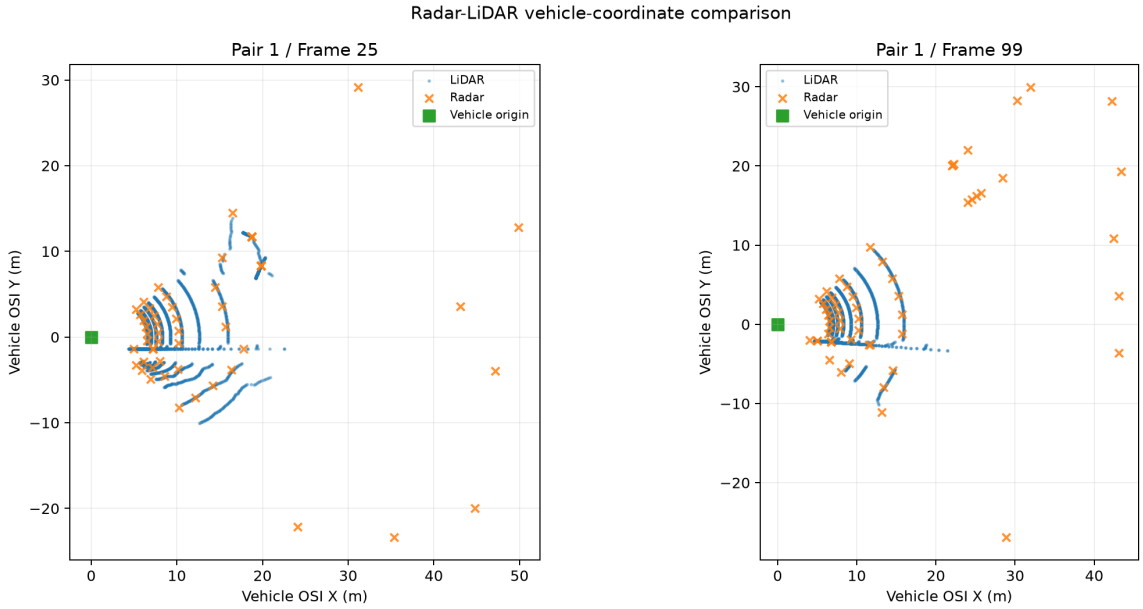


Figure 1. task23_05d1_pair1_radar_lidar_overlay.png - Radar and LiDAR sample overlay in vehicle relative OSI frame; large nearest-point outliers in some cases prohibit a blanket matching claim.

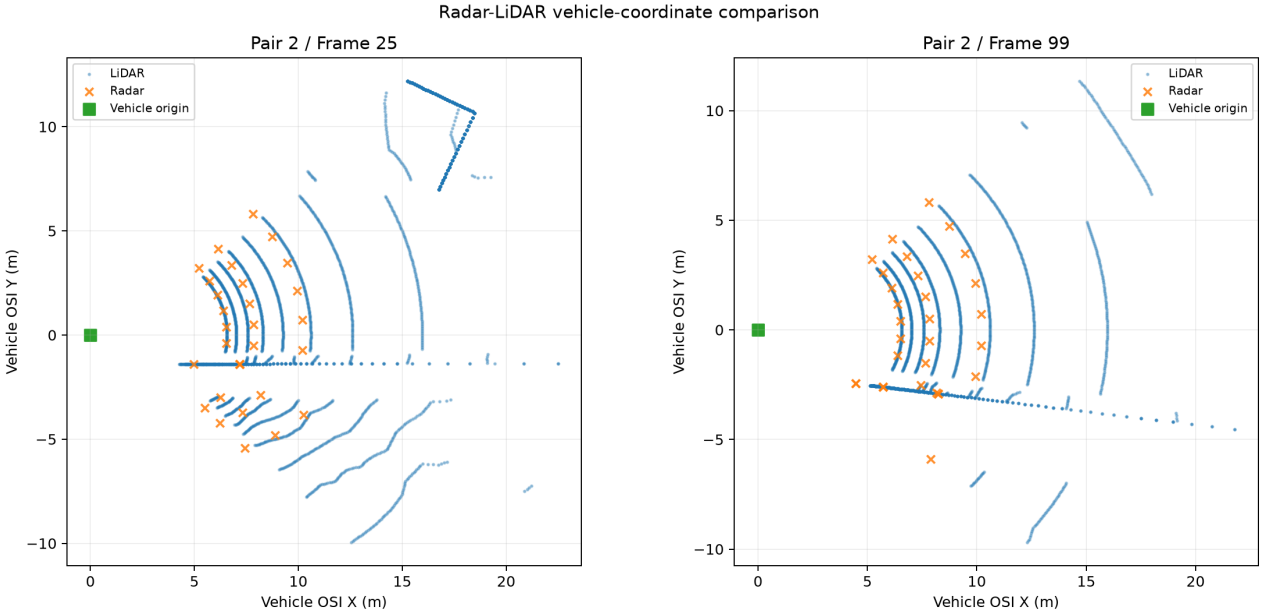


Figure 2. task23_05d1_pair2_radar_lidar_overlay.png - Alternative sampled geometry; compare density, coverage and nearest-point artifacts rather than assuming correspondences.

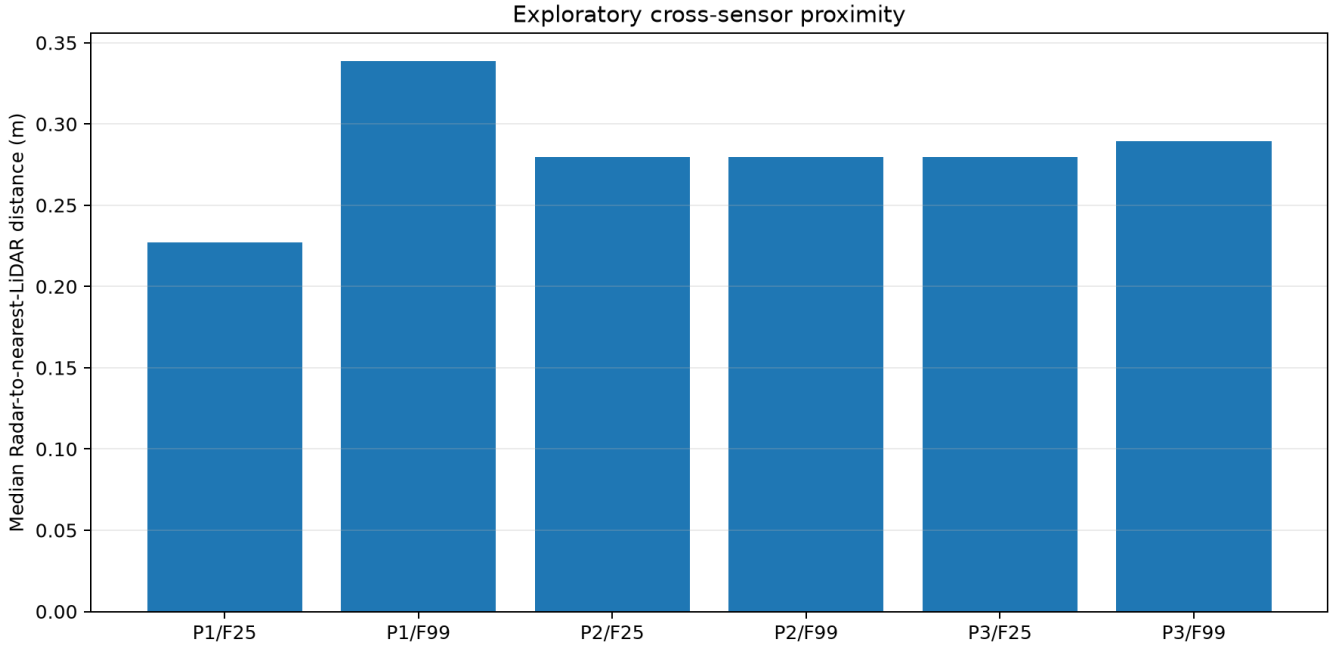


Figure 3. task23_05d1_nearest_distance_summary.png - Nearest point-distance summary. Some P95 values are high; object association is unverified.

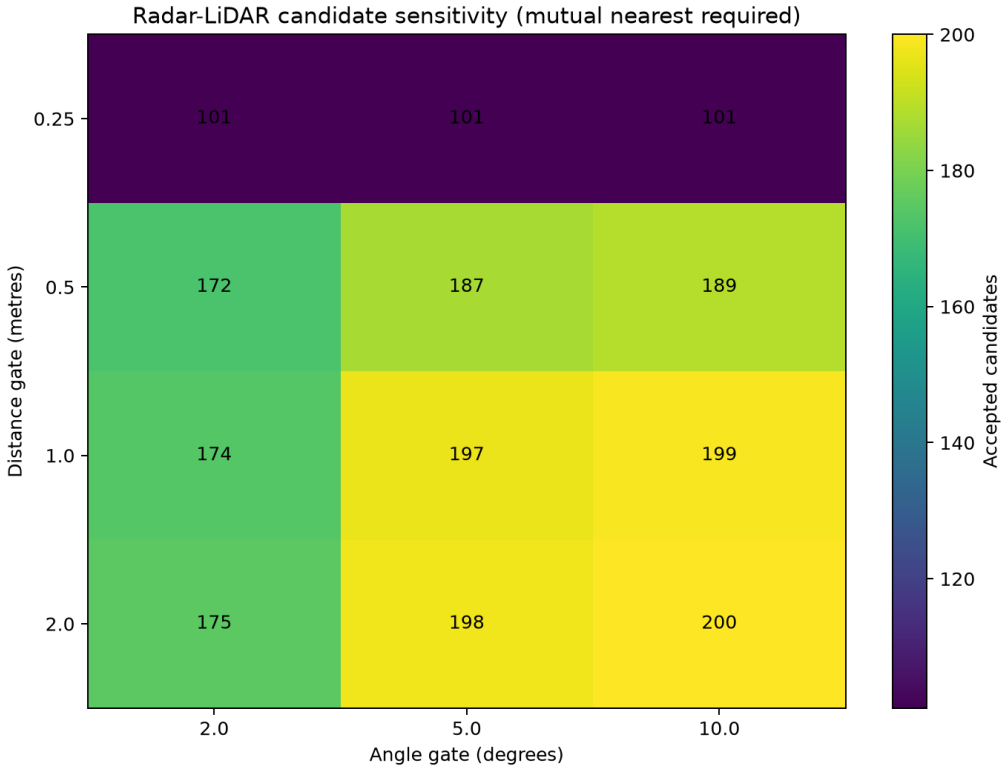


Figure 4. task23_05d6_threshold_sensitivity.png - Sensitivity of provisional association candidates to thresholds, not object detection accuracy.

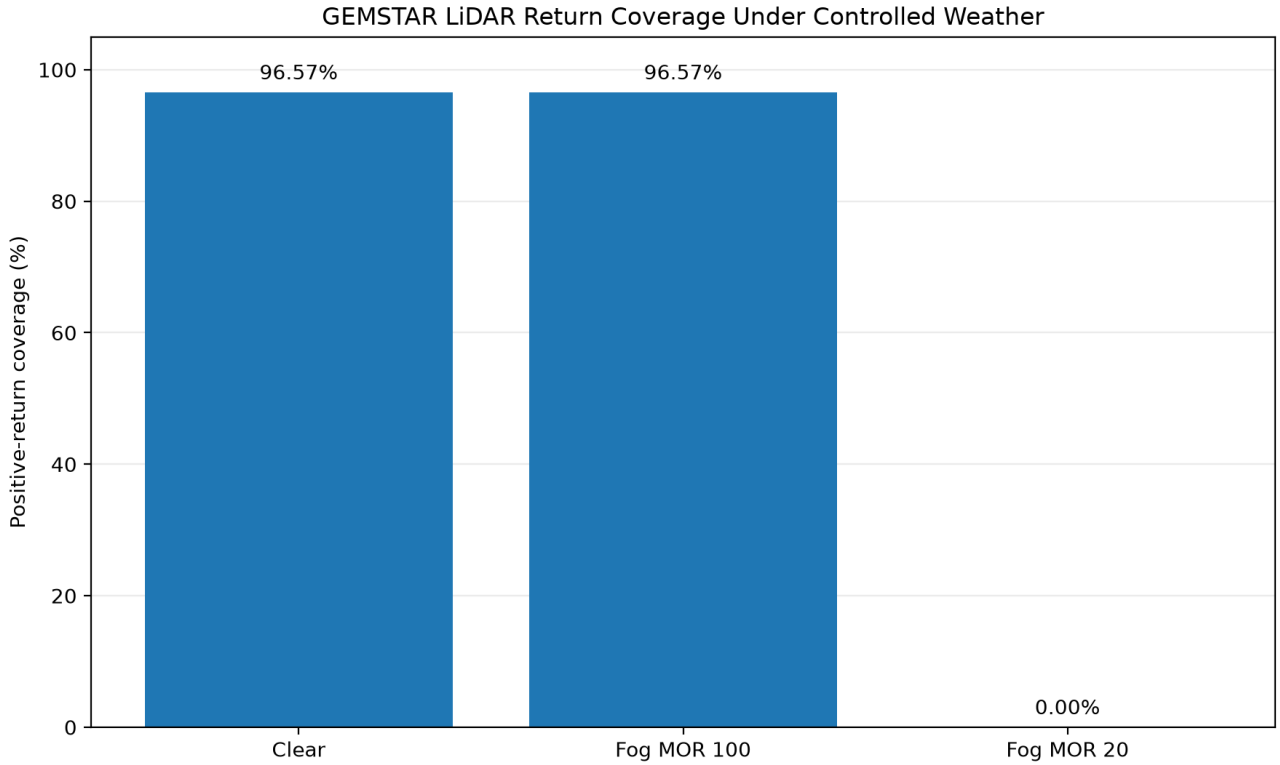


Figure 5. task23_11E_lidar_coverage_by_weather.png - Controlled LiDAR weather availability, useful as a future scheduler input only when validated against perception quality.

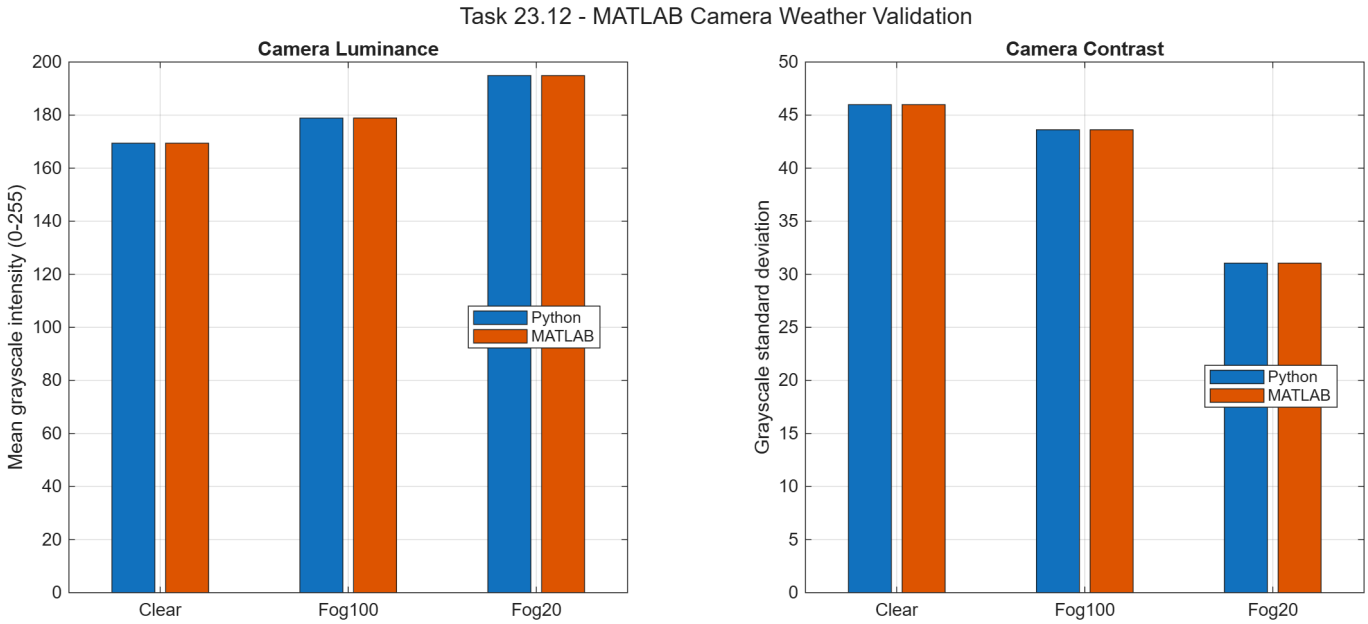


Figure 6. task23_12C_07_matlab_camera_weather_comparison.png - Selected Camera image statistics under static weather scenarios; not fusion accuracy.

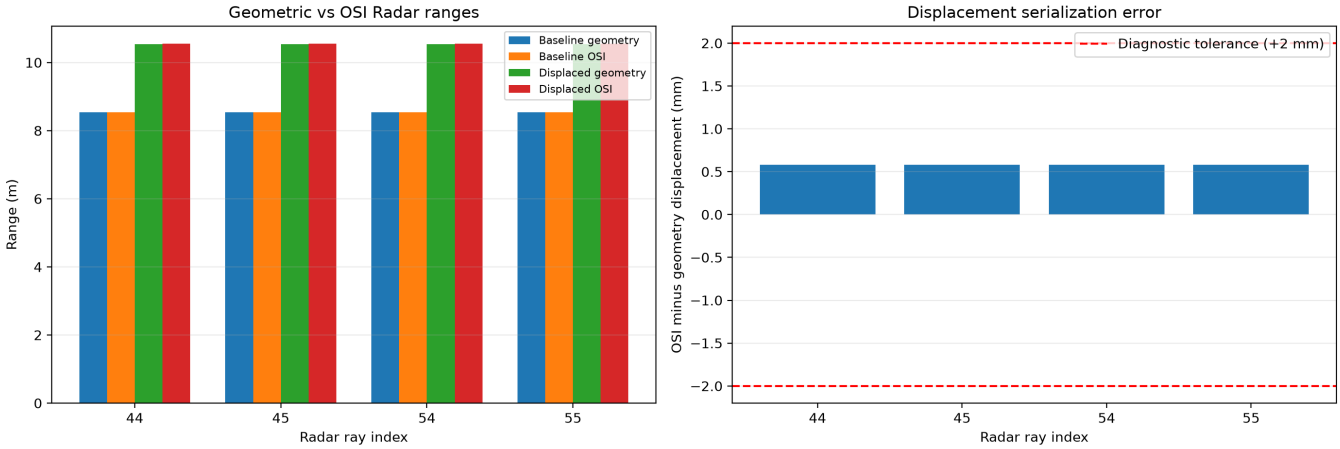
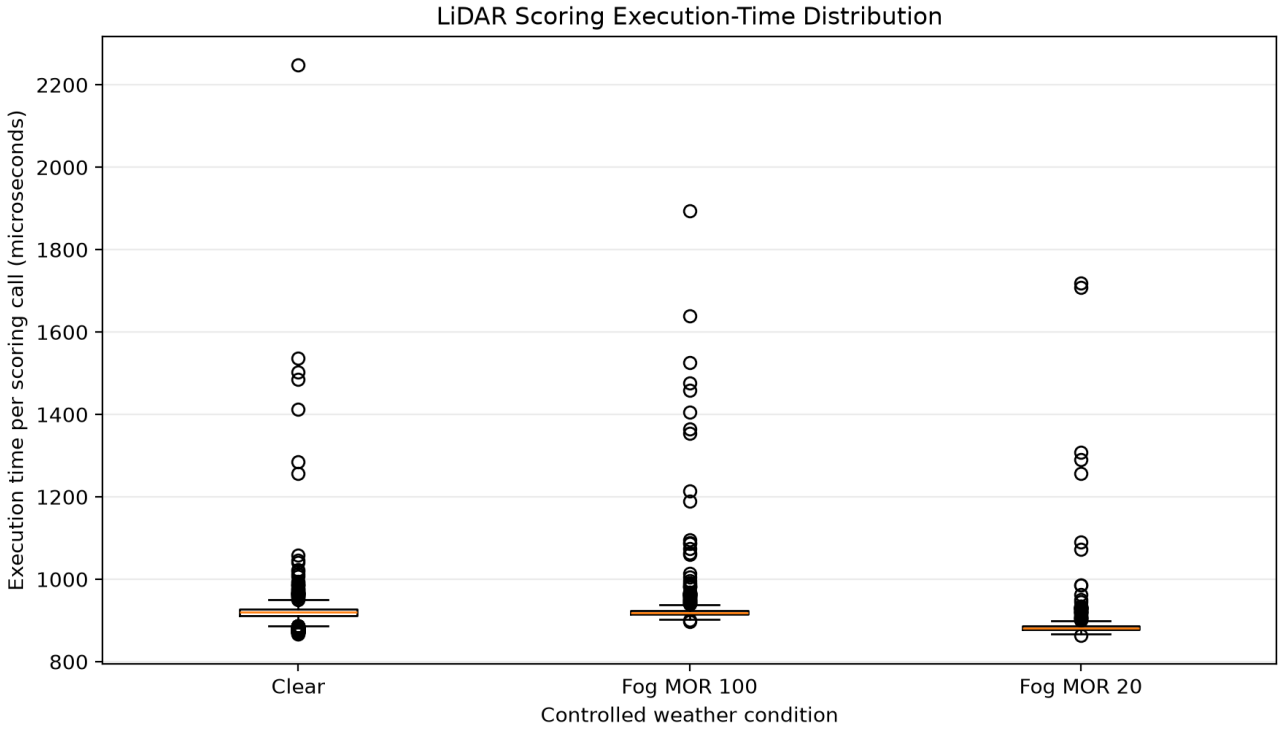


Figure 7. task23_13C_02_radar_serialization_comparison.png - Static Radar geometry-to-OSI preservation of calibration displacement.



Python on PC; 600 repeated calls per fixed frame, not ARM-FPGA timing

Figure 8. task23_11G_02B_timing_distribution.png - Desktop Python LiDAR score duration only; hardware board measurements still to come.

Cross-sensor architecture and case-study figures

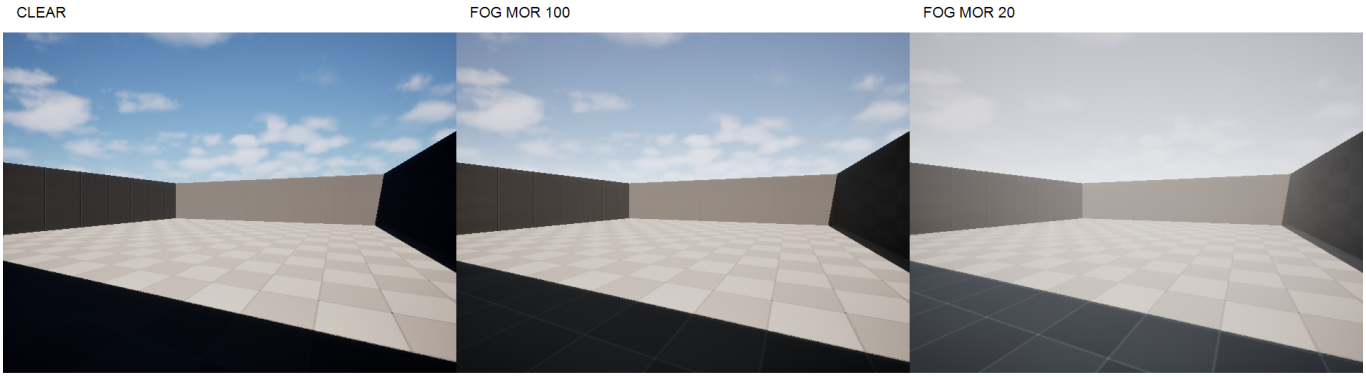


Figure 9. task23_12C_03_three_weather_comparison.png - Illustrative static comparison across three independent Camera weather runs; no same-run three-sensor synchronization implied.

A rigorous evidence chain for the thesis

Step 1 - source recording: immutable OSI SensorView and GroundTruth pair, runtime configuration hash, weather and map provenance. Step 2 - parsing: sensor payload type, ID, size and timestamp semantics. Step 3 - sensor-specific preprocessing: quality features, valid/invalid/no-return distinctions, range or 3D coordinates with explicit frame. Step 4 - fusion prerequisites: same-run time offsets, verified 6-DOF extrinsics and Camera projection residuals. Step 5 - association and EKF: ground-truth object tracking, quality thresholds and failure modes. Step 6 - scheduler: matched CPU/FPGA energy and p99 latency comparisons subject to preselected quality constraints.

What MATLAB adds, and what it does not

MATLAB scripts already exist for initial Camera verification, LiDAR scan validation, Radar 3D parity, and controlled Camera weather luminance/contrast. These form a cross-tool verification channel. A useful enhancement is one multi-panel consolidated dashboard per modality, explicitly sourced from existing CSVs; MATLAB analysis cannot supply missing independent sensor calibration, synchronized data, or measured energy.

Detailed subtask and code inventory

The table below is organized by archived script name and research objective. A filename identifies work that exists; it is not by itself a claim that every planned experiment or verification was completed.

Task 0 - source scripts (12)

task23_07af_measure_radar_ranges.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_07ag_geometry_validation.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_07ah_move_target.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_07ai_displacement_validation.py - Controlled change to target geometry for calibration.

task23_07aj_full_radar_comparison.py - Compare two measurements/scenarios using specified method.

task23_07ak_map_audit.py - Inspect source, data structure, indexing, or evidence assumptions.

task23_07ap_trace_audit.py - Inspect source, data structure, indexing, or evidence assumptions.

task23_07ar_move_and_verify.py - Read-only verification of scenario artifacts, data validity, or configuration.

task23_07ar_move_persistent.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_07ar_verify_persistent.py - Read-only verification of scenario artifacts, data validity, or configuration.

task23_07ar_verify_saved_map.py - Read-only verification of scenario artifacts, data validity, or configuration.

task23_07as_finalize_radar.py - Consolidate evidence and create a research-ready result.

Task 02 - source scripts (1)

task23_02_inspect_osi_fields.py - Inspect recorded or simulated payloads before accepting analysis.

Task 03a - source scripts (1)

task23_03a_camera_format_inspection.py - Inspect recorded or simulated payloads before accepting analysis.

Task 03b - source scripts (1)

task23_03b_camera_decode_candidates.py - Interpret encoded Camera/Radar/OSI measurements.

Task 03c - source scripts (1)

task23_03c_verify_color_format.py - Read-only verification of scenario artifacts, data validity, or configuration.

Task 03d - source scripts (1)

task23_03d_camera_quality_analysis.py - Calculate image/sensor-quality metrics rather than detection accuracy.

Task 03e - source scripts (1)

task23_03e_matlab_camera_validation.m - MATLAB cross-tool calculation/plot based on exported research data.

Task 04a - source scripts (1)

task23_04a_lidar_inspection.py - Inspect recorded or simulated payloads before accepting analysis.

Task 04b - source scripts (1)

task23_04b_lidar_range_reconstruction.py - Reconstruct geometry, range, or Cartesian coordinates.

Task 04c - source scripts (2)

task23_04c1_export_lidar_raw_for_matlab.py - MATLAB cross-tool calculation/plot based on exported research data.

task23_04c2_matlab_lidar_validation.m - MATLAB cross-tool calculation/plot based on exported research data.

Task 04d - source scripts (3)

task23_04d1_lidar_ray_index_audit.py - Inspect source, data structure, indexing, or evidence assumptions.

task23_04d2_lidar_source_audit.py - Inspect source, data structure, indexing, or evidence assumptions.

task23_04d4_correct_lidar_geometry.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

Task 04e - source scripts (10)

task23_04e1_groundtruth_pose_inspection.py - Inspect recorded or simulated payloads before accepting analysis.

task23_04e2_host_pose_recovery.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_04e3_lidar_vehicle_transform.py - Apply or check reference-frame transformations.

task23_04e4d_finalize_vehicle_coordinates.py - Audit or finalize coordinate frame and geometry conventions.

task23_04e5a_world_transform.py - Apply or check reference-frame transformations.

task23_04e5b_groundtruth_geometry_audit.py - Inspect source, data structure, indexing, or evidence assumptions.

task23_04e5c_host_object_association.py - Explore correspondence candidates and rejection or identity assumptions.

task23_04e5d_bounding_box_consistency.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_04e5e_lidar_object_id_audit.py - Inspect source, data structure, indexing, or evidence assumptions.

task23_04e5f_actor_spatial_consistency.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

Task 05a - source scripts (1)

task23_05a_radar_configuration_audit.py - Inspect source, data structure, indexing, or evidence assumptions.

Task 05c - source scripts (4)

task23_05c1_radar_measurements.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_05c2_radar_startup_audit.py - Inspect source, data structure, indexing, or evidence assumptions.

task23_05c3_radar_3d_reconstruction.py - Reconstruct geometry, range, or Cartesian coordinates.

task23_05c4_matlab_radar_verification.m - MATLAB cross-tool calculation/plot based on exported research data.

Task 05d - source scripts (6)

task23_05d1_radar_lidar_alignment.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_05d2_radar_lidar_timestamp_audit.py - Inspect source, data structure, indexing, or evidence assumptions.

task23_05d3_candidate_associations.py - Explore correspondence candidates and rejection or identity assumptions.

task23_05d4_actor_association_audit.py - Inspect source, data structure, indexing, or evidence assumptions.

task23_05d5_association_failure_analysis.py - Explore correspondence candidates and rejection or identity assumptions.

task23_05d6_association_robustness.py - Explore correspondence candidates and rejection or identity assumptions.

Task 06d - source scripts (2)

task23_06d8i_remove_radar_tester.py - Run defined behavior or invalid-input checks.

task23_06d8j_verify_radar_actors.py - Read-only verification of scenario artifacts, data validity, or configuration.

Task 07 - source scripts (1)

task23_07_automated_calibration.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

Task 07g - source scripts (1)

task23_07g_inspect_sensor_managers.py - Inspect recorded or simulated payloads before accepting analysis.

Task 07r - source scripts (1)

task23_07r_validate_osi.py - Test or numerical validation of a prior dataset/implementation.

Task 07s - source scripts (2)

task23_07s_decode_radar.py - Interpret encoded Camera/Radar/OSI measurements.

task23_07s_decode_radar_before_fix.py - Interpret encoded Camera/Radar/OSI measurements.

Task 07u - source scripts (1)

task23_07u_reflection_inspection.py - Inspect recorded or simulated payloads before accepting analysis.

Task 07v - source scripts (1)

task23_07v_radar_configuration.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

Task 07w - source scripts (1)

task23_07w_reference_geometry.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

Task 07y - source scripts (1)

task23_07y_align_radar_target.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

Task 08 - source scripts (11)

task23_08_clear_set_map.py - Modify or prepare experimental map (historical operation; do not rerun).

task23_08_clear_verify_map.py - Read-only verification of scenario artifacts, data validity, or configuration.

task23_08_compare_clear_fog.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_08_displace_target.py - Controlled change to target geometry for calibration.

task23_08_finalize_lidar.py - Consolidate evidence and create a research-ready result.

task23_08_inspect_lidar_osi.py - Inspect recorded or simulated payloads before accepting analysis.

task23_08_inspect_world_config.py - Inspect recorded or simulated payloads before accepting analysis.

task23_08_set_map_config.py - Modify or prepare experimental map (historical operation; do not rerun).

task23_08_validate_displacement.py - Test or numerical validation of a prior dataset/implementation.

task23_08_verify_displacement.py - Read-only verification of scenario artifacts, data validity, or configuration.

task23_08_verify_map_config.py - Read-only verification of scenario artifacts, data validity, or configuration.

Task 09 - source scripts (9)

task23_09_auto_cube_measure.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_09_camera_set_map.py - Modify or prepare experimental map (historical operation; do not rerun).

task23_09_camera_verify_map.py - Read-only verification of scenario artifacts, data validity, or configuration.

task23_09_compare_camera_images.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_09_decode_camera_frames.py - Interpret encoded Camera/Radar/OSI measurements.

task23_09_displace_camera_target.py - Controlled change to target geometry for calibration.

task23_09_finalize_camera.py - Consolidate evidence and create a research-ready result.

task23_09_measure_camera_cube.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_09_verify_camera_displacement.py - Read-only verification of scenario artifacts, data validity, or configuration.

Task 10 - source scripts (5)

task23_10_inspect_map_reference.py - Inspect recorded or simulated payloads before accepting analysis.

task23_10_set_fog_map.py - Prepare controlled weather configuration (historical operation).

task23_10_set_weather_map.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_10_verify_fog_map.py - Read-only verification of scenario artifacts, data validity, or configuration.

task23_10_verify_weather_map.py - Read-only verification of scenario artifacts, data validity, or configuration.

Task 11B - source scripts (5)

task23_11B_displaced_clear_sensorview.py - Controlled change to target geometry for calibration.

task23_11B_full_frame_integrity.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_11B_inspect_actual_timestamps.py - Inspect recorded or simulated payloads before accepting analysis.

task23_11B_inspect_timestamp_schema.py - Inspect recorded or simulated payloads before accepting analysis.

task23_11B_preliminary_fog_sensorview.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

Task 11C - source scripts (3)

task23_11C_inspect_npz.py - Inspect recorded or simulated payloads before accepting analysis.

task23_11C_validate_lidar_masks.py - Test or numerical validation of a prior dataset/implementation.

task23_11C_validate_npz.py - Test or numerical validation of a prior dataset/implementation.

Task 11D - source scripts (4)

task23_11D_baseline_graphs.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_11D_baseline_stability.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_11D_clear_baseline.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_11D_compare_raw_frames.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

Task 11E - source scripts (12)

task23_11E_clear_distance_thresholds.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_11E_compare_first_reflections.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_11E_full_clear_variability.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_11E_generate_weather_graphs.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_11E_sample_clear_osi.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_11E_sample_clear_osi_fixed.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_11E_set_clear_map.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_11E_set_fog100_map.py - Prepare controlled weather configuration (historical operation).

task23_11E_set_fog20_map.py - Prepare controlled weather configuration (historical operation).

task23_11E_verify_clear_map.py - Read-only verification of scenario artifacts, data validity, or configuration.

task23_11E_verify_fog100_map.py - Read-only verification of scenario artifacts, data validity, or configuration.

task23_11E_verify_fog20_map.py - Read-only verification of scenario artifacts, data validity, or configuration.

Task 11F - source scripts (11)

task23_11F_03B_test_reliability.py - Calculate positive-return availability or handle invalid inputs.

task23_11F_04A_verify_real_osi.py - Read-only verification of scenario artifacts, data validity, or configuration.

task23_11F_04B_full_reliability.py - Calculate positive-return availability or handle invalid inputs.

task23_11F_05A_validate_reliability_csv.py - Test or numerical validation of a prior dataset/implementation.

task23_11F_05B_weather_score_validation.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_11F_06A_abnormal_reflection_tests.py - Run defined behavior or invalid-input checks.

task23_11F_06B2_malformed_assertions.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_11F_06B_missing_malformed_tests.py - Run defined behavior or invalid-input checks.

task23_11F_07A_weather_visualization.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_11F_07B_frame_reliability_graph.py - Calculate positive-return availability or handle invalid inputs.

task23_11F_lidar_reliability.py - Calculate positive-return availability or handle invalid inputs.

Task 11G - source scripts (3)

task23_11G_01_preliminary_profile.py - Development-PC compute workload diagnostic, not energy measurement.

task23_11G_02A_timing_distribution.py - Evaluate intervals or processing durations under stated sampling conditions.

task23_11G_02B_timing_analysis.py - Evaluate intervals or processing durations under stated sampling conditions.

Task 12A - source scripts (4)

task23_12A_04C_camera_weather_payload_audit.py - Inspect source, data structure, indexing, or evidence assumptions.

task23_12A_set_camera_clear_map.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_12A_verify_camera_clear_map.py - Read-only verification of scenario artifacts, data validity, or configuration.

task23_12A_verify_clear_map_evidence.py - Read-only verification of scenario artifacts, data validity, or configuration.

Task 12B - source scripts (5)

task23_12B_set_camera_fog100_map.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_12B_set_camera_fog20_map.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_12B_verify_clear_camera_payload.py - Read-only verification of scenario artifacts, data validity, or configuration.

task23_12B_verify_fog100_map.py - Read-only verification of scenario artifacts, data validity, or configuration.

task23_12B_verify_fog20_map.py - Read-only verification of scenario artifacts, data validity, or configuration.

Task 12C - source scripts (7)

task23_12C_01_camera_timing_audit.py - Inspect source, data structure, indexing, or evidence assumptions.

task23_12C_02_extract_matched_camera_images.py - Extract selected frames, sensor values, or metrics.

task23_12C_05_camera_quality_metrics.py - Calculate image/sensor-quality metrics rather than detection accuracy.

task23_12C_06_camera_cpu_preprocessing.py - Implement sensor-specific preprocessing baseline.

task23_12C_07_matlab_camera_validation.m - MATLAB cross-tool calculation/plot based on exported research data.

task23_12C_07_run_matlab_and_plot.m - MATLAB cross-tool calculation/plot based on exported research data.

task23_12C_07_run_matlab_and_plot_original.m - MATLAB cross-tool calculation/plot based on exported research data.

Task 12D - source scripts (2)

task23_12D_01_camera_robustness_tests.py - Run defined behavior or invalid-input checks.

task23_12D_02B_verify_before_restore.py - Read-only verification of scenario artifacts, data validity, or configuration.

Task 13B - source scripts (2)

task23_13B_02_decode_radar_calibration.py - Interpret encoded Camera/Radar/OSI measurements.

task23_13B_03_validate_radar_serialization.py - Test or numerical validation of a prior dataset/implementation.

Task 13C - source scripts (2)

task23_13C_01_radar_robustness_tests.py - Run defined behavior or invalid-input checks.

task23_13C_02_radar_comparison_plot.py - Generate visual research evidence from previously recorded data.

Task 13D - source scripts (1)

task23_13D_01_finalize_radar.py - Consolidate evidence and create a research-ready result.

Task 14B - source scripts (1)

task23_14B_03F_camera_intrinsic_sanity_check.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

Task 15 - source scripts (6)

task23_15_camera_adapter.py - Wrap validated sensor-specific operation into common interface.

task23_15_coordinate_adapter.py - Wrap validated sensor-specific operation into common interface.

task23_15_lidar_adapter.py - Wrap validated sensor-specific operation into common interface.

task23_15_radar_adapter.py - Wrap validated sensor-specific operation into common interface.

task23_15_timestamp_bundle.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_15_unified_preprocessing.py - Implement sensor-specific preprocessing baseline.

CSV evidence and reproducibility register

CSV file	Rows	Example fields
task23_05d1_radar_lidar_alignment.csv	6	PairID, Frame, RadarPoints...
task23_05d2_radar_lidar_timestamps.csv	15	PairID, Frame, TargetFrame...
task23_05d2_timestamp_progression.csv	9	PairID, FromFrame, ToFrame...
task23_05d3_candidate_association_details.csv	237	PairID, Frame, RadarRayIndex...
task23_05d3_candidate_association_summary.csv	6	PairID, Frame, RadarPoints...
task23_05d4_actor_association_details.csv	237	PairID, Frame, RadarRayIndex...
task23_05d4_actor_association_summary.csv	6	PairID, Frame, RadarPoints...

CSV file	Rows	Example fields
task23_05d4_actor_candidate_counts.csv	20	PairID, Frame, LiDARActorID...
task23_05d5_association_classification.csv	237	PairID, Frame, RadarRayIndex...
task23_05d5_candidate_actor_distribution.csv	20	PairID, Frame, AssignedNearestLiDARActorID...
task23_05d5_failure_cases.csv	44	PairID, Frame, RadarRayIndex...
task23_05d5_frame_summary.csv	6	PairID, Frame, RadarReturns...
task23_05d6_frame_actor_concentration.csv	6	PairID, Frame, RadarReturns...
task23_05d6_threshold_sensitivity.csv	24	DistanceGateM, AngleGateDeg, RequireMutual...
task23_11G_01_preliminary_timing.csv	3	weather, repetitions, reflection_slots...
task23_11G_02A_raw_timing_distribution.csv	1800	round, weather, repetition...
task23_11G_02B_timing_statistics.csv	3	weather, samples, mean_us...
task23_12C_01_camera_timing_audit.csv	30	scenario, frame, timestamp_seconds...
task23_12C_02_matched_camera_frames.csv	3	scenario, frame_index, timestamp_seconds...
task23_12C_05_camera_quality_metrics.csv	6	scenario, region, mean_luminance...

CSV file	Rows	Example fields
task23_12C_07_matlab_camera_validation.csv	3	Scenario, MATLAB_Luminance, Python_Luminance...
task23_13B_02_radar_calibration_ranges.csv	80	Scenario, FileName, Frame...
task23_13B_03_radar_serialization_comparison.csv	4	Ray, BaselineGeometryM, BaselineOSIM...
task23_13C_01_radar_robustness_tests.csv	12	Test, ExpectedAccept, ActualAccept...
task23_14B_03F_camera_intrinsic_sanity_check.csv	2	Condition, DistanceM, CubeFaceWidthM...
task23_14_readiness_summary.csv	8	Component, Evidence, Status
task23_15_preprocessing_acceptance.csv	10	Component, Tests, Passed...

Missing figures, why they matter, and how to create them

Figure / experiment	Why it is needed	Source / feasibility
One full recorded-OSI 3-sensor data-flow figure	Proves real payloads traverse integrated adapters, beyond synthetic examples	Needs raw same-run OSI and measured frame-ID/timestamp trace.
Per-sensor capture-time jitter/offset plot	Validates temporal compatibility for moving-object fusion	Requires instrumenting sensor capture events or justified clock semantics.

Figure / experiment	Why it is needed	Source / feasibility
Camera reprojection overlays and pixel-error histogram	Tests Camera-Radar/LiDAR extrinsics and object correspondence	Requires known 3D/2D matching targets in one validated run.
Dynamic object GT association/EKF panels	Enables scientifically defensible fusion tracking claims	Task 24: GT matches, trajectories and failure analysis.
CPU vs FPGA energy per correct perception figure	This is the main hardware research claim	Later ARM/FPGA power instrumentation and matched workload benchmarking.
Combined data-flow architectural diagram	Clarifies how OSI, adapters, status, fusion and scheduler connect	Can be authored now, clearly labeling unimplemented stages.

Scientific interpretation and handover

The checked results are a reproducible software and simulator preprocessing baseline. They do not establish real-world accuracy, dynamic multi-object perception, CPU-versus-FPGA energy improvement, or measured deadlines. Those claims require separate matched scenarios, explicit ground-truth evaluation, and physical hardware benchmarking.

All embedded plots/images are original archived figures except where a figure is explicitly described as a newly synthesized summary. The archived source file name accompanies each caption. Raw multi-gigabyte OSI sources are not embedded and were not reprocessed for these PDFs.